Homework Assignment 3:
Compressible Flow

Mech 588 — Spring, 2023

Due Date: April 3

There are several places in the notes where I cheerfully assert that something is true without any proof. This question will close some of those gaps.

1. (3 marks) For the analytic flux for 1D compressible, inviscid flow, show that

$$F = \frac{\partial F}{\partial U} U$$

2. (5 marks) For Roe's scheme, with left and right states U_L and U_R, respectively, show that

$$F_L - F_R = \tilde{A} \cdot (U_L - U_R)$$

3. (7 marks) Suppose you were writing a program to solve compressible, inviscid flow problems and you wanted to use implicit time advance. Not only would you need to be able to calculate the flux Jacobians, but you would need to assemble them into a block tridiagonal matrix.
   For a one-dimensional, uniform mesh with spacing Δx, write out the fully-discrete system you would use for Roe's flux-difference splitting. Be sure to use second-order spatial accuracy for the flux integral. Use the implicit Euler time advance scheme. When you compute the left-hand side matrix, use only first-order spatial accuracy (in other words, compute the Jacobians, etc, for the first-order accurate spatial discretization. You may neglect variations in the analytic flux Jacobian A (and its parts). Feel free to simply write A or $\tilde{A}$ for the Jacobians, and to define expressions like $U_{i+\frac{1}{2};i}$ to abbreviate the extrapolated solution at the interface.

1) To start, we should non-dimensionalize our equation.

Remember, we have an initial equation

$$\frac{\partial U}{\partial t} + \frac{\partial f}{\partial x} = 0$$

where $U = \begin{pmatrix} \rho \\ \rho u \\ E_t \end{pmatrix}$, $f = \begin{pmatrix} \rho u \\ \rho u^2 + p \\ u(E_t + p) \end{pmatrix}$

with relations as follows,

$$p = \rho R T, \quad E_t = \rho C_v T + \rho \frac{u^2}{2}$$

Knowing $R = C_p - C_v$ & $\gamma = C_p / C_v$,

we can define the following non-dimensionalizations which we can relate to the freestream conditions,

$$\rho = \rho^* \rho_\infty, \quad x = x^* L, \quad u = u^* C_\infty, \quad t = \frac{t^* L}{C_\infty},$$

$$p = p^* \rho_\infty C_\infty^2, \quad T = T^* \frac{C_\infty^2}{\gamma R}$$

We also need non-dimensionalizations of some relations

Pressure can be given by $p^* = \frac{\rho^* T^*}{\gamma}$

Speed of sound is given by $c^* = \sqrt{T^*} = \sqrt{\frac{\gamma p^*}{\rho^*}}$

Heat flux is given by $\bar{q}^* = -\frac{k c_\infty^2}{\gamma R L} \nabla^* T^*$

Finally we need to derive a non dimensionalization for E_t^*.

Given $E_t = \rho C_v T + \frac{\rho u^2}{2}$, we can plug the non dimensionalizations above to yield the following

$$E_t = \rho^* \rho_\infty C_v \frac{T^* c_\infty^2}{\gamma R} + \rho^* \rho_\infty \frac{(u^* c_\infty)^2}{2}$$

$$= \rho^* \rho_\infty c_\infty^2 \left[\frac{C_v T^*}{\gamma R} + \frac{u^{*2}}{2} \right]$$

knowing $\gamma = C_p / C_v$, & $R = C_p - C_v$ we can get

$$= \rho^* \rho_\infty c_\infty^2 \left[\frac{C_v T^*}{(C_p/C_v)(C_p - C_v)} + \frac{u^{*2}}{2} \right]$$

$$= \rho^* \rho_\infty c_\infty^2 \left[\frac{C_v T^*}{\frac{C_p^2}{C_v} - C_p} + \frac{u^{*2}}{2} \right]$$

multiply top & bottom by $\frac{1}{C_v}$

$$= \rho^* \rho_\infty c_\infty^2 \left[\frac{C_v T^* \cdot \frac{1}{C_v}}{\left(\frac{C_p^2}{C_v} - C_p\right) \cdot \frac{1}{C_v}} + \frac{u^{*2}}{2} \right]$$

$$= \rho^* \rho_\infty c_\infty^2 \left[\frac{T^*}{\frac{C_p^2}{C_v^2} - \frac{C_p}{C_v}} + \frac{u^{*2}}{2} \right]$$

$$= \rho^* \rho_\infty c_\infty^2 \left[\frac{T^*}{\frac{c_p}{c_v}\left(\frac{c_p}{c_v}-1\right)} + \frac{u^{*2}}{2} \right]$$

$$\gamma$$

$$E_t^* = \rho^* \rho_\infty c_\infty^2 \left[\frac{T^*}{\gamma(\gamma-1)} + \frac{u^{*2}}{2} \right]$$

Remember, we started with $\frac{\partial U}{\partial t} + \frac{\partial f}{\partial x} = 0$

where $U = \begin{pmatrix} \rho \\ \rho u \\ E_t \end{pmatrix}$, $f = \begin{pmatrix} \rho u \\ \rho u^2 + p \\ u(E_t + p) \end{pmatrix}$

which gives 3 equations

$$\frac{\partial \rho}{\partial t} + \frac{\partial \rho u}{\partial x} = 0$$

$$\frac{\partial \rho u}{\partial t} + \frac{\partial \rho u^2}{\partial x} + \frac{\partial p}{\partial x} = 0$$

$$\frac{\partial E_t}{\partial t} + \frac{\partial u(E_t + p)}{\partial x} = 0$$

We can nondimensionalize these one at a time

Now that we have all the nondimensionalizations, we arrive at the following nondimensional equations

$$\rho_\infty \frac{c_\infty}{L}\left(\frac{\partial \rho^*}{\partial t^*} + \frac{\partial \rho^* u^*}{\partial x^*} \right) = 0$$

$$\rho_\infty \frac{c_\infty^2}{L}\left(\frac{\partial \rho^* u^*}{\partial t^*} + \frac{\partial \rho^* u^{*2}}{\partial x^*} \right) = 0$$

$$\rho_\infty \frac{c_\infty^3}{L}\left(\frac{\partial E_t^*}{\partial t^*} + \frac{\partial (E_t^* + p^*)}{\partial x^*} \right) = 0$$

We notice these are the same equations as the beginning but with scaling factors. Lets move to the problem

$$U = \begin{pmatrix} \rho \\ \rho u \\ E_t \end{pmatrix}, \quad f = \begin{pmatrix} \rho u \\ \rho u^2 + p \\ u(E_t + p) \end{pmatrix}$$

Show $f = \frac{\partial f}{\partial U} U$

We can rewrite f in terms of u. for example we notice $f_1 = u_2$

We can write p as a function of ρ, ρu & E_t.

Remember, $p = \rho R T$, we know $R = c_p - c_v$ & $\gamma = c_p / c_v$

$$p = \rho(c_p - c_v)T, \quad E_t = \rho c_v T + \rho \frac{u^2}{2}$$

We also know $U_1 = \rho$, $U_2 = \rho u$, $U_3 = E_t$

from this we can get $\frac{U_2}{U_1} = \frac{\rho u}{\rho} = u$

We can rewrite T in terms of ρ, c_v, E_t & u

$$E_t - \frac{1}{2}\rho u^2 = \rho c_v T \rightarrow T = \frac{E_t}{\rho c_v} - \frac{u^2}{2c_v}$$

Plugging this into the equation for p gives

$$p = \rho R T = \rho(c_p - c_v)T = \rho c_p T - \rho c_v T$$

$$p = \rho C_p \left(\frac{E_t}{\rho C_v} - \frac{u^2}{2C_v} \right) - \rho C_v \left(\frac{E_t}{\rho C_v} - \frac{u^2}{2C_v} \right)$$

$$p = \frac{\rho C_p E_t}{\rho C_v} - \frac{\rho C_p u^2}{2C_v} - \frac{\rho E_t}{\rho} + \frac{\rho u^2}{2}$$

$$p = \gamma E_t - \gamma \left(\frac{1}{2} \right) \rho u^2 - E_t + \frac{1}{2} \rho u^2$$

$$p = \gamma E_t - \gamma \left(\frac{1}{2} \right) (\rho u) u - E_t + \left(\frac{1}{2} \right) (\rho u) u$$

$E_t = U_3$, $\rho u = U_2$, $u = \frac{U_2}{U_1}$, $E_t = U_3$, $\rho u = U_2$, $u = U_2/U_1$

$$p = \gamma U_3 - \gamma \left(\frac{1}{2} \right) U_2 \frac{U_2}{U_1} - U_3 + \frac{1}{2} U_2 \frac{U_2}{U_1}$$

$$p = \gamma U_3 - \left(\frac{1}{2} \right) \gamma \frac{U_2^2}{U_1} - U_3 + \left(\frac{1}{2} \right) \frac{U_2^2}{U_1}$$

$$p = \left(\frac{1}{2} \right) (1 - \gamma) \frac{U_2^2}{U_1} + (\gamma - 1) U_3$$

Remember $f = \begin{pmatrix} \rho u \\ \rho u^2 + p \\ u(E_t + p) \end{pmatrix}$, knowing $U_1 = \rho$, $U_2 = \rho u$, $U_3 = E_t$, $\frac{U_2}{U_1} = u$, we can rewrite f

$$f = \begin{pmatrix} U_2 \\ \frac{U_2}{U_1} U_2 + \frac{1}{2}(1 - \gamma) \frac{U_2^2}{U_1} + (\gamma - 1) U_3 \\ \frac{U_2}{U_1} \left(U_3 + \frac{1}{2}(1 - \gamma) \frac{U_2^2}{U_1} + (\gamma - 1) U_3 \right) \end{pmatrix}$$

$$F = \begin{pmatrix} U_2 \\ \frac{U_2}{U_1} U_2 + \frac{1}{2}(1-\gamma)\frac{U_2^2}{U_1} + (\gamma-1)U_3 \\ \frac{U_2}{U_1}\left(U_3 + \frac{1}{2}(1-\gamma)\frac{U_2^2}{U_1} + (\gamma-1)U_3\right) \end{pmatrix}$$

$$F = \begin{pmatrix} U_2 \\ \frac{U_2^2}{U_1} + \frac{1}{2}(1-\gamma)\frac{U_2^2}{U_1} + (\gamma-1)U_3 \\ \frac{U_2}{U_1}\left((1+\gamma-1)U_3 + \frac{1}{2}(1-\gamma)\frac{U_2^2}{U_1}\right) \end{pmatrix}$$

$$F(U) = \begin{pmatrix} U_2 \\ \frac{1}{2}(3-\gamma)\frac{U_2^2}{U_1} + (\gamma-1)U_3 \\ \gamma\frac{U_2 U_3}{U_1} + \frac{1}{2}(\gamma-1)\frac{U_2^3}{U_1^2} \end{pmatrix}$$

Now we can determine $\frac{\partial F}{\partial U}$

We know that given two vectors $\vec{x}$, $\vec{y}$

Where $\vec{y}$ has length m, & $\vec{x}$ has length n, we get $\frac{\partial \vec{y}}{\partial \vec{x}} = \begin{pmatrix} \frac{\partial y_1}{\partial x_1} & \frac{\partial y_1}{\partial x_2} & \cdots & \frac{\partial y_1}{\partial x_n} \\ \frac{\partial y_2}{\partial x_1} & \frac{\partial y_2}{\partial x_2} & \cdots & \frac{\partial y_2}{\partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial y_m}{\partial x_1} & \frac{\partial y_m}{\partial x_2} & \cdots & \frac{\partial y_m}{\partial x_n} \end{pmatrix}$

Thus, we can compute $A = \frac{\partial f}{\partial v}$

$$A = \frac{\partial f}{\partial v} = \begin{pmatrix} \frac{\partial f_1}{\partial v_1} & \frac{\partial f_1}{\partial v_2} & \frac{\partial f_1}{\partial v_3} \\ \frac{\partial f_2}{\partial v_1} & \frac{\partial f_2}{\partial v_2} & \frac{\partial f_2}{\partial v_3} \\ \frac{\partial f_3}{\partial v_1} & \frac{\partial f_3}{\partial v_2} & \frac{\partial f_3}{\partial v_3} \end{pmatrix}$$

$$\frac{\partial f_1}{\partial v_1} = \frac{\partial}{\partial v_1}(v_2) = 0, \quad \frac{\partial f_1}{\partial v_2} = \frac{\partial}{\partial v_2} v_2 = 1, \quad \frac{\partial f_1}{\partial v_3} = \frac{\partial}{\partial v_3} v_2 = 0$$

$$\frac{\partial f_2}{\partial v_1} = \frac{\partial}{\partial v_1} \frac{1}{2}(3-\gamma)\frac{v_2^2}{v_1} + (\gamma-1)v_3$$

$$= \frac{1}{2}(3-\gamma) v_2^2 \frac{\partial}{\partial v_1} v_1^{-1} = \frac{1}{2}(3-\gamma) v_2^2 (-v_1^{-2}) = -\frac{1}{2}(3-\gamma)\left(\frac{v_2}{v_1}\right)^2$$

$$\frac{\partial f_2}{\partial v_2} = \frac{\partial}{\partial v_2} \frac{1}{2}(3-\gamma)\frac{v_2^2}{v_1} + (\gamma-1)v_3 = \frac{1}{2}(3-\gamma)\frac{1}{v_1}\frac{\partial}{\partial v_2} v_2^2$$

$$= \frac{1}{2}(3-\gamma)\frac{1}{v_1}(2v_2) = (3-\gamma)\frac{v_2}{v_1}$$

$$\frac{\partial f_2}{\partial v_3} = \frac{\partial}{\partial v_3} \frac{1}{2}(3-\gamma)\frac{v_2^2}{v_1} + (\gamma-1)v_3 = (\gamma-1)\frac{\partial}{\partial v_3} v_3 = \gamma-1$$

$$\frac{\partial f_3}{\partial v_1} = \frac{\partial}{\partial v_1} \gamma \frac{v_3 v_2}{v_1} + \frac{1}{2}(\gamma-1)\frac{v_2^3}{v_1^2}$$

$$= \gamma v_3 v_2 \frac{\partial}{\partial v_1} v_1^{-1} + \frac{1}{2}(\gamma-1) v_2^3 \frac{\partial}{\partial v_1} v_1^{-2}$$

$$= \gamma v_3 v_2 (-v_1^{-2}) + \frac{1}{2}(\gamma-1) v_2^3 (-2 v_1^{-3})$$

$$\frac{\partial F_3}{\partial v_1} = -\gamma \frac{v_3 v_2}{v_1^2} - (\gamma-1)\left(\frac{v_2}{v_1}\right)^3$$

$$\frac{\partial f_3}{\partial v_2} = \frac{\partial}{\partial v_2}\gamma \frac{v_3 v_2}{v_1} + \frac{1}{2}(\gamma-1)\frac{v_2^3}{v_1^2} = \gamma \frac{v_3}{v_1}\frac{\partial}{\partial v_2}\cancel{v_2}^{1} + \frac{1}{2}(\gamma-1)\frac{1}{v_1^2}\frac{\partial}{\partial v_2} v_2^3$$

$$= \gamma \frac{v_3}{v_1} + \frac{1}{2}(\gamma-1)\frac{1}{v_1^2}(3v_2^2) = \gamma \frac{v_3}{v_1} + \frac{3}{2}(\gamma-1)\frac{v_2^2}{v_1^2}$$

$$\frac{\partial f_3}{\partial v_3} = \frac{\partial}{\partial v_3}\gamma \frac{v_3 v_2}{v_1} + \frac{1}{2}\cancel{(\gamma-1)}^{0}\frac{v_2^3}{v_1^2} = \gamma \frac{v_2}{v_1}\frac{\partial}{\partial v_3}\cancel{v_3}^{1} = \gamma \frac{v_2}{v_1}$$

$$A = \frac{\partial f}{\partial v} = \begin{pmatrix} 0 & 1 & 0 \\ -\frac{1}{2}(3-\gamma)\left(\frac{v_2}{v_1}\right)^2 & (3-\gamma)\frac{v_2}{v_1} & \gamma-1 \\ -\gamma\frac{v_3 v_2}{v_1^2} - (\gamma-1)\left(\frac{v_2}{v_1}\right)^3 & \gamma\frac{v_3}{v_1} + \frac{3}{2}(\gamma-1)\frac{v_2^2}{v_1^2} & \gamma\frac{v_2}{v_1} \end{pmatrix}$$

$$\frac{\partial f}{\partial v} v = \begin{pmatrix} 0 & 1 & 0 \\ -\frac{1}{2}(3-\gamma)\left(\frac{v_2}{v_1}\right)^2 & (3-\gamma)\frac{v_2}{v_1} & \gamma-1 \\ -\gamma\frac{v_3 v_2}{v_1^2} - (\gamma-1)\left(\frac{v_2}{v_1}\right)^3 & \gamma\frac{v_3}{v_1} + \frac{3}{2}(\gamma-1)\frac{v_2^2}{v_1^2} & \gamma\frac{v_2}{v_1} \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix}$$

1st equation $= 0(v_1) + 1(v_2) + 0(v_3) = v_2$

2nd equation $= -\frac{1}{2}(3-\gamma)\left(\frac{v_2}{v_1}\right)^2 v_1 + (3-\gamma)\frac{v_2}{v_1}(v_2) + (\gamma-1)(v_3)$

$$= -\frac{1}{2}(3-\gamma)\frac{v_2^2}{v_1} + (3-\gamma)\frac{v_2^2}{v_1} + (\gamma-1)v_3$$

$$= \left(1-\frac{1}{2}\right)(3-\gamma)\frac{U_2^2}{U_1} + (\gamma-1)U_3$$

$$= \frac{1}{2}(3-\gamma)\frac{U_2^2}{U_1} + (\gamma-1)U_3$$

3rd equation $= \left(-\gamma\frac{U_3U_2}{U_1^2} - (\gamma-1)\left(\frac{U_2}{U_1}\right)^3\right)(U_1) + \left(\gamma\frac{U_3}{U_1} + \frac{3}{2}(\gamma-1)\frac{U_2^2}{U_1^2}\right)(U_2) + \gamma\frac{U_2}{U_1}(U_3)$

$$= -\gamma\frac{U_3U_2}{U_1} - (\gamma-1)\frac{U_2^3}{U_1^2} + \gamma\frac{U_2U_3}{U_1} + \frac{3}{2}(\gamma-1)\frac{U_2^3}{U_1^2} + \gamma\frac{U_2U_3}{U_1}$$

$$= \gamma\frac{U_2U_3}{U_1} + \left(\frac{3}{2}-\frac{2}{2}\right)(\gamma-1)\frac{U_2^3}{U_1^2}$$

$$= \gamma\frac{U_2U_3}{U_1} + \frac{1}{2}(\gamma-1)\frac{U_2^3}{U_1^2}$$

$$\frac{\partial F}{\partial U}U = \begin{pmatrix} U_2 \\ \frac{1}{2}(3-\gamma)\frac{U_2^2}{U_1} + (\gamma-1)U_3 \\ \gamma\frac{U_2U_3}{U_1} + \frac{1}{2}(\gamma-1)\frac{U_2^3}{U_1^2} \end{pmatrix} = F$$